ANSWER KEY

The Real Number System

Answer each question, then check your work.

Name _____ Date _____ Period _____

SCORE *3.0* Basic Understanding

1.1 Converting Rational Numbers

Convert each number as directed. Write fractions in simplest form.

1. Write as a decimal: 5/8

= **0.625**

e.g. 0.5

Solution:

5 ÷ 8 = 0.625. It terminates, so it is rational.

2. Write as a fraction: 0.75

= **3/4**

simplest form, e.g. 3/4

Solution:

0.75 = 75/100 = 3/4 in simplest form.

1.2 Real Number Classification

Classify the number as Rational (R) or Irrational (IR).

3. Classify: √169

Classification: **R**

type R or IR

Solution:

√169 = 13, a whole number, so it is rational.

SCORE 4.0 Target Content

1.1 Converting Rational Numbers

Convert each number as directed. Write fractions in simplest form.

4. Write as a fraction: $0.\overline{3}$ (the 3 repeats)

= **1/3**

simplest form

Solution:

One digit repeats → put it over 9: 3/9 = 1/3.

5. Write as a decimal: $2\frac{1}{3}$ (2 and 1/3)

= **2.333...**

use ... for the repeating part

Solution:

2 1/3 = 2 + 1/3 = 2.333... (the 3 repeats).

1.2 Real Number Classification

Classify the number as Rational (R) or Irrational (IR).

6. Classify: $\sqrt{50}$

`Classification:` **IR**

type R or IR

Solution:

50 is not a perfect square, so √50 = 5√2 ≈ 7.07... — non-terminating and non-repeating, so irrational.

Simplify each expression first, then classify the result.

7. Classify: $\sqrt{2} \times \sqrt{8}$

`Classification:` **R**

simplify, then R or IR

Solution:

√2 × √8 = √16 = 4, a whole number, so rational.

8. Classify: $\sqrt{75} \div \sqrt{3}$

Classification: **R**

simplify, then R or IR

Solution:

√75 ÷ √3 = √25 = 5, a whole number, so rational.

1.3 Estimating Non-Perfect Square Roots

Estimate each value to the *nearest tenth* by clicking the number line.

9. Estimate: $\sqrt{28}$



Solution:

$5^2 = 25$ and $6^2 = 36$, so √28 ≈ 5.29, which rounds to 5.3.

10. Estimate: $\sqrt{60}$



Solution:

$7^2 = 49$ and $8^2 = 64$, so √60 ≈ 7.746..., which rounds to 7.7.

SCORE *5.0* Advanced Content

1.2 Real Number Classification

Evaluate or simplify each expression if possible, then classify as Rational (R) or Irrational (IR).

11. Classify: $2 + \sqrt{16}$

Classification: **R**

simplify, then R or IR

Solution:

√16 = 4, so 2 + √16 = 6, an integer, so rational.

12. Classify: $5 + \sqrt{7}$

Classification: **IR**

type R or IR

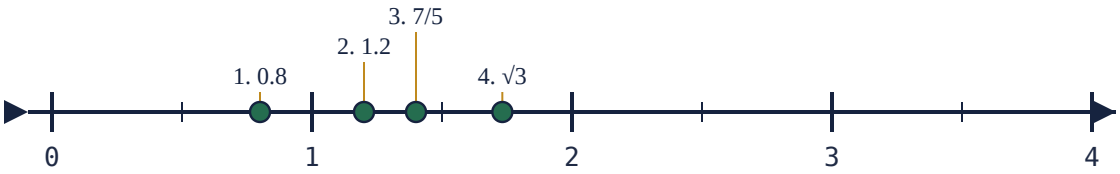
Solution:

√7 is irrational, and a rational number (5) plus an irrational number is irrational.

1.4 Ordering Real Numbers

Click the values in order from *least to greatest*. Each one drops onto the number line.

13. Set: { 0.8, √3, 1.2, 7/5 }



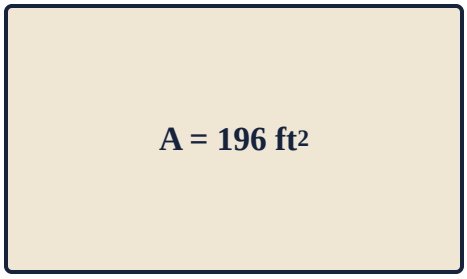
Order: **0.8, 1.2, 7/5, √3**

Solution:

As decimals: 0.8, √3 ≈ 1.73, 1.2, 7/5 = 1.4 → least to greatest: 0.8, 1.2, 7/5, √3.

1.5 Area

A square garden has an area of 196 ft². Find the side length and the perimeter.



14. Find the side length and perimeter of the square garden.

Side length = **14 ft**

Perimeter = **56 ft**

Solution:

Side = √196 = 14 ft. Perimeter = 4 × 14 = 56 ft.